

# Analysis of RTO and RPO of a Service Stored on Amazon Web Service (AWS) and Google Cloud Engine (GCE)

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**Abstract**— Today, the availability of the application is beyond everything. Application can not be accessed even in a minute will become major problem for the business and threat the reputation of the company especially that need 24/7 availability of application. Companies that have traditional physical environments typically must duplicate their infrastructure to ensure the availability of spare capacity in the event of a disaster. Nowadays, there are many cloud providers that offer service called DRaaS (Disaster Recovery as a Service) to facilitate company needs in case of disaster recovery for their system to ensure their business continuity. Many companies still confuse which provider that can suit their system. Since RTO and RPO is the most critical two metrics in disaster recovery planning. This paper introduces implementation design method and analysis of two metrics between two cloud providers.

**Keywords**—DRaaS; AWS; Google Cloud;

## I. INTRODUCTION

The emergence of cloud computing technology has changed the paradigm of the business world in applying technology in its business processes massively. The earlier cloud services based on pay-per-use models, IaaS (Infrastructure as a Service), PaaS(Platform as a Service), and SaaS(Software as a Service) have made many businesses previously using traditional infrastructure system moved to cloud-based infrastructure to improve the reliability of their business based. In their system usually include many applications that become core business of the company.

Today, the availability of the application is beyond everything. Application can not be accessed even in a minute will become major problem for the business and threat the reputation of the company especially that need 24/7 availability of application. Disaster is one of many reason that system or application can be unavailable to deliver service. Fires, water main breaks, hurricanes, floods, telecom outages, even hardware failures, software bugs, administrator error results in data corruption or deleted objects could be happened considered as disaster could happen in daily basis [1]. If

company's system including all applications and database are affected and provoke downtime, this may be the substansial problem even downfall for the business.

Disaster recovery planning came up as one of many attempts that can save our system from undesirable disaster. Amazon Web Service (AWS) [2] define disaster recovery is anything in respect of preparing for and recovering from a disaster. Everything that has bad effects on a company's business sustainability could be termed a disaster. This belonging hardware or software failure, a network outage, a power outage, physical damage to a building like fire or flooding, human error, or some other significant event[2].

RTO(Recovery Time Objective) and RPO(Recovery Point Objective) is the most critical two metrics in disaster recovery planning[3]. RTO value means the duration of service unavailable until recover and start the service again. In the other hand, RPO means maximum amount of data that can be lost when restoration is successful in time. A successful disaster recovery key consists of maintain the service always online according to SLA (Service Level Agreement) also meeting RPO and PTO goals[4].

Cloud computing establish high reliability of service for preserving sensitive and important client data[5]. According to Assante et al [6] utilizing latest trend of technology that is cloud computing for companies is one of strategic skills to maintain the companies existance between competitors.

Today, there are lot of DRaaS provider such as AWS, Google Cloud, OVH that offered disaster recovery in their service. Every provider almost offered same features that confusing customers. Therefore we conduct several testing between providers that focus on RTO and RPO values to help customer for seeking the best choice for their disaster recovery planning.

## II. BACKGROUND

Disaster Recovery is an effort to preparing and recovering from a disaster. Everything that has bad effects on a company's business sustainability could be termed a disaster. This belonging hardware or software failure, a network outage, a power outage, physical damage to a building like fire or flooding, human error, or some other significant event. Every company must have its own infrastructure to support the business process especially that needs strong demand of application availability. Therefore, disaster recovery plan is not an option but it is a must.

Many advantages could be achieve applying disaster recovery plan, in-between assist minimize defect of critical functions, recover operations quickly and successfully when system is down, assure stability of organization, also preserves organization's asset [7]. Moreover, disaster recovery plan permit organization or company to state higher availability and reliability of services than other competitor [8].

There are three approaches to build disaster recovery, hot site, cold site, and warm site. Hot site means that companies have exact duplicate of their entire architecture system including server, data, storage, and network called secondary location that will not be affected by any event affected the primary location. The data is replicated between primary and secondary site. It could be active-active replication, where all databases are sync each other and replicated in two-way direction or active-passive replication, when data is replicated in one way direction. Meanwhile, Cold site is the type of approach that just send backup file of the system. Warm site denote average way between cold and hot.

In disaster recovery terms, the mechanism to switch service from primary site to secondary site when the service failed due to disaster is called failover[9]. In failover state, all services including application and data will be handled by secondary site. Otherwise, when the primary site problem has been resolved and the service is switch back to the primary site its called failback.

Today, building a disaster recovery system in traditional architecture has so much disadvantages such as complexity and cost including operating, maintaining, human resource and building cost[10]. Disaster Recovery as a Service emerge to overcome cost and complexity problem in physical infrastructure.

Disaster Recovery as a Service is cloud based service responsible to ensure system availability and recovery from disaster. DRaaS brought different way to back up substantial system including application, data also resources and quickly recover systems after a disaster with less cost and complexity. The advantages of implementing DRaaS are Availability, Cost Reduction, Simplicity, Visibility, Scalability, Flexibility. Moreover we are allow to conducting DR testing in case the disaster is really happen as a trial.

In order to achive well-prepared disaster recovery plan, several metrics must be set. Podaras et al.[11] explain various parameter in recovery test i.e Critical Business Function (CBF), Maximum Acceptable Outage (MAO); Recovery Time Objectives (RTOs), and Business Impact Analysis (BIA). In

the other hand, Wood et al[12] specify other metrics such as RPO, Performance, Consistency, Geographic Separation.

In this research we will use two metrics to analysis our recovery disaster testing between providers. RTO stands for Recovery Time Objectives means the duration of service for example an application unavailable until recover and start the application service again. The less time required, the faster the recoveries will need to be.

In the other hand, RPO means maximum amount of data that can be lost when restoration is successful in time. RPO declares the data might be lost during disaster, the lower the RPO, the higher the total cost to save primary infrastructure environment for recovery [12]. In order to decrease the amount of data lost, perform a frequent backup scheduling might be the best choice. Figure 1 depicts a very clear distinction between RTO and RPO.

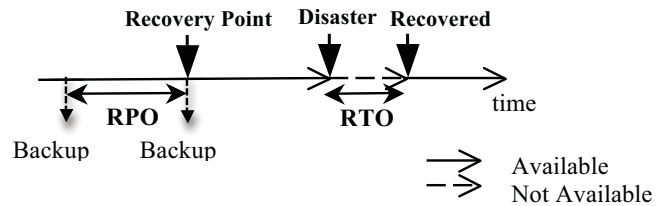


Fig. 1. The distinguish between RTO and RPO[12]

Amazon Web Service and Google Cloud Engine are two of hundreds provider serve disaster recovery as a service in one of their products. AWS(Amazon Web Service) offered services and features to meet customer requirement in their disaster recovery plan including storage, server, networking, database, deployment orchestration equipped with interactive tool[13]. Since AWS have multiple region site around the world, customer could choose the most appropriate location site.

AWS have several approaches to implement disaster recovery strategy. There are 4 possible approaches i.e Backup & Restore, Pilot Light, Warm Standby, and multisite. Every approach has own advantages and disadvantages regarding customer business process needs. Backup and Restore option allowed us to backup data in our system and sent to Amazon S3, cloud storage provided by Amazon Web Service, through the network. Pilot light option is similar to previous backup and restore option, with minimal version of environment running on the cloud. Pilot light uses data mirroring replication to ensure the consistency of data.

Warm standby is extend infrastructure of pilot light version. We have duplicate system on AWS that is always running thus decreases recovery time. Load balancer play a role as failover mechanism between on-premise and AWS. The last one is multi site option that offered smallest RTO and RPO values that is real time failover system on the cloud. This option supported by active-active replication of database.

Google Compute Engine also have solutions model for Disaster Recovery as a Service especially for application

recovery. Hot standby, Warm standby and Cold standby are solution offered for applications running on Google Cloud Engine[13].

### III. PROPOSED METHOD

The method used in this research in adapting Hosam's data centre recovery phase. The recovery phase is divided in 3 sub-phase. The first phase is disaster recovery assessment plan. The next phase is disaster recovery action, and the last is system and application testing, also the analysis after testing.

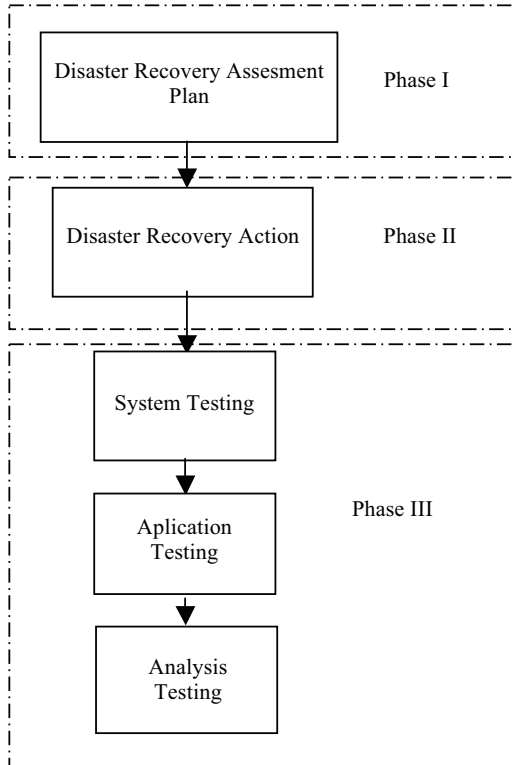


Fig. 2. Implementation Phases

Figure 2 explain the disaster recovery plan of this research. Disaster recovery assessment plan contains all activities about design system and application will be testing in this research, also design procedures for entire steps. The main metrics for evaluating disaster recovery will be set and define in this phase along with testing and evaluating procedures scenario.

The second phase, disaster recovery action is the phase where all of the plans and designs will be executing. Deploying secondary system architecture, verify the operating systems and communication software between two sites are working properly also implement backup procedures and failover method will be done in this phase.

System and application testing helps to verify that the recovery procedures work as expected. Several testing will be repeated to ensure reliability of the system. Analysis phase will be held when all procedures testing are done and will be release suggestion for improvement when necessary.

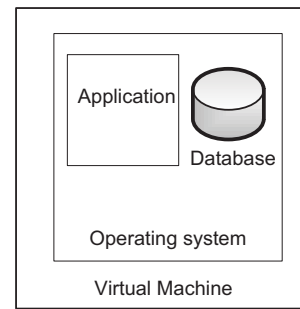


Fig. 3. System application logical tier

Figure 3 explain application testing environment used in this research. Application run inside operating system that lied in virtual machine environment since the primary server infrastructure is located in the cloud dedicated server. Since the the main concern of disaster recovery is preserving critical data and system especially application that run core business of the company. We use database replication to maintain consistency between primary and secondary system. We will use Ubuntu 16.04 LTS as operating system, web-based application, and MySQL as database server in three different cloud systems. However virtual machine type might vary in each cloud system since we will use three different cloud providers as implementation testing.

Testing scenario in this disaster recovery implementation is divided in 2 scenario. The first testing scenario is between CloudKilat as primary server then Amazon Web Service as secondary server. The consistency data between two system was guaranteed by master-slave replication. As normal activity user will be directed to primary server, when disaster is occured system will be automatically switched to secondary server. This mecanism called failover that ensure the availability of system, therefore user client still could access the service

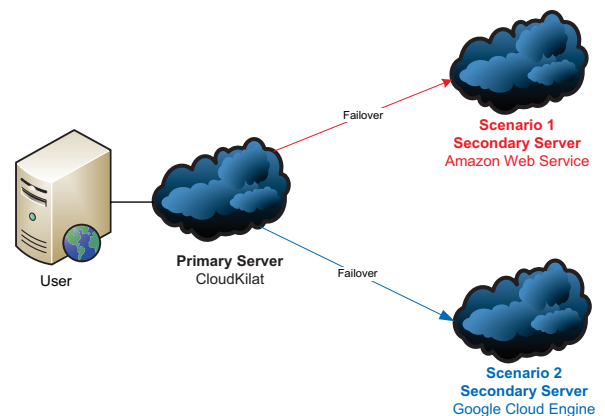


Fig. 4. Implementation Scenario

The second scenario involved another provider that is Google Cloud Engine as secondary system. The workflow of

the system is not much different from another one except different provider that act as secondary server. Furthermore, failover tools used might vary in each scenario since every cloud provider has its own architecture system.

The location of primary and secondary server are different in region and zone. CloudKilat located in Indonesia, while Amazon Web Service and Google Cloud Engine located in United States.

#### IV. ANALYSIS

In this chapter we analyze several previous study that related with our research particularly on RTO and RPO metrics in experimental testing. Sindhura[14] conducted experimental study between offline tape-based and cloud backup based DRaaS. Tape-based means the mechanism of system backup is saved on tape storage device transferred through LAN media. Otherwise, cloud-based backup transferred via internet connection.

TABLE I RTO and RPO define values[14]

| Type of Backup | RTO           | RPO      |
|----------------|---------------|----------|
| Tape Based     | 5 to 24 hours | 12 hours |
| Cloud Based    | 0-4 hours     | 1 hours  |

Table 1 represents a very significant different between two model of backup. RTO value in tape-based backup depending on the size of data needed for restoration. Moreover, RPO value in cloud based backup is less than tape-based since the periode of data backup more often.

Suguna[3] classify seven tiers of disaster recovery in relationship with RTO and RPO as revealed in Table 2. The seventh tier is the best choice for lower RTO and RPO values, yet the cost prepared would be increased.

TABLE II Disaster Recovery Level[3]

| Tier | Description                 | RTO       | RPO      |
|------|-----------------------------|-----------|----------|
| 1    | Point in time tape backup   | 2-7 days  | 2-24 hrs |
| 2    | Tape backup to remote site  | 1-3 days  | 2-24 hrs |
| 3    | Disk point in time copy     | 2-24 hrs  | 2-24 hrs |
| 4    | Remote logging              | 12-24 hrs | 5-30 hrs |
| 5    | Concurrent ReEX             | 1-12 hrs  | 5-30 min |
| 6    | Mirrored data               | 1-4 hrs   | 0-5 min  |
| 7    | Mirrored data with failover | 0-60 min  | 0-5 min  |

As seen in Table 2, this research will implement mirrored data with failover as model for infrastructure. Therefore, the RTO and RPO values for this research are 0-60 minutes and 0-5 minutes range respectively. Since in this research cloud server location might be quite influential, we will consider location as additional parameter that affect RTO and RPO values.

#### V. DISCUSSION AND CONCLUSION

This research proposed new approach to consider when comparing DRaaS providers in RTO and RPO perspective. Since in previous research several criteria used to contrass DRaaS provider were price, infrastructure, orchestration, location, etc. Comparing cloud to cloud scenario of disaster recovery between different provider is not exist yet, therefore we propose implementation method also RTO and RPO benchmarking analysis between providers as reference for choosing cloud providers. Since this research will build mirrored data with failover based DRaaS, we have stated RTO value is 0-60 minutes and RPO value is 0-5 minutes.

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